HydroDyn implementation plan:

Hydrodynamic loads on strip-theory members at a shallow angle relative to the free surface

1 Background

The previous HydroDyn strip-theory implementation could not model the hydrodynamic loads on partially submerged sections accurately. While the hydrostatic load is computed by integrating the hydrostatic pressure over the instantaneous wetted surface (MHstLMod=2) and is accurate for arbitrary member-free surface intersections, the hydrodynamic load components were still simplified. More specifically, if the centerline point of a section was below the free surface with wave stretching or below the still water level (SWL) without wave stretching, the section was modeled as fully submerged. If the centerline point was out of water, the whole section was treated as dry. As a side effect, this simplification effectively treated the waterplane of the member as always perpendicular to the centerline, regardless of member inclination and wave slope. This was especially problematic for nearly horizontal members and/or steep waves, because the Froude-Krylov component could become inconsistent with the more exact hydrostatic component.

To address this limitation with hydrodynamic loads and to close the fidelity gap with hydrostatic loads, we develop a modified hydrodynamic load model for circular members close to or intersecting the free surface at a shallow angle. We focus on members with circular sections only. An equivalent treatment for rectangular sections, if possible, will likely be significantly more complex and is not currently planned.

The modified model is intended to work in parallel with the existing formulation suitable for members at a steeper angle relative to the free surface. A smooth blending from one model to the other is implemented based on the local section orientation relative to the free surface with wave stretching or the SWL if there is no wave stretching. If the angle between the section normal and the free surface normal is between 65 and 115 degrees, the new hydrodynamic load model is used exclusively. If the angle is below 55 or above 125 degrees, the existing model is used exclusively. A smooth blending between the two is implemented for the gap regions. These angle ranges are intentionally selected so that the transition gap regions avoid 45-degree diagonal supports prevalent in offshore structure designs. By performing the blending on a nodal level as opposed to on a member level, the algorithm takes into account the local wave slope and member section orientation without assuming rigid and straight members.

2 Hydrodynamic load model

The detailed formulation of each hydrodynamic load component along with some implementation details is provided below. These include distributed transverse loads, distributed axial loads for tapered members, and joint loads. For the first two load components, we exclusively focus on circular cross sections. The joint hydrodynamic loads are computed based on an abstract joint reference area, volume, and orientation without a clear distinction of cross-section geometries. This is necessary because members with different cross sections can connect to the same joint. Therefore, the joint load modification shall be applied regardless of section geometries.

2.1 Depth and submergence of a circular section

Before discussing the hydrodynamic load formulation, it is helpful to first establish the geometry of the problem. Figure 1 shows a circular member section with a section local xyz-coordinate system. The local z-axis is along the member centerline perpendicular to the section. Without loss of generality, the section local x-axis is parallel to the earth-fixed horizontal XY-plane, and the local y-axis follows from the right-hand convention. We further define unit vectors $\vec{\imath}$, $\vec{\jmath}$, and $\vec{k}$ along the section local x-, y-, and z-directions.

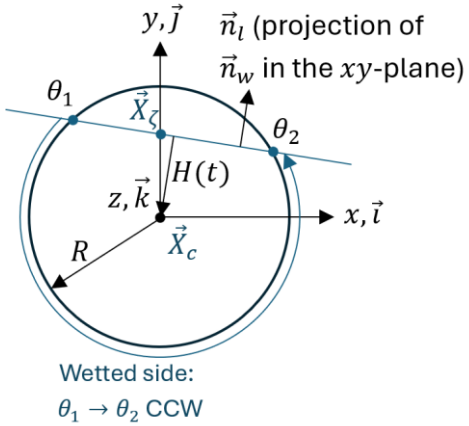


Figure 1. A partially submerged circular section.

For a surface-piercing section or a section just beneath the free surface, the section added masses in the heave and sway directions are strongly dependent on the submergence of the section. Typically, a center depth H is used as the key parameter. In the canonical potential-flow problem of a vertical circular section (horizontal cylinder) interacting with a horizontal linearized free surface, H is straightforwardly defined as the vertical distance from the still water level to the center of the circle measured downward. Therefore, $H = -R$ where R is the radius of the circle implies the circle is out of water with its lowest point

tangent to the SWL. Conversely, $H = R$ means the circle is fully submerged with its top tangent to the SWL. For a deeply submerged section, $H/R \to +\infty$.

In a nonlinear time-domain solver like OpenFAST, we need to generalize H to support sections inclined from vertical and an instantaneous free surface that deviates from the SWL with a nonzero wave slope. The most straightforward approach is to compute H in the section plane based on the normal distance from the section center, located at $\vec{X}_c = (X_c, Y_c, Z_c)$ in the earth-fixed system, to the in-plane cut of the instantaneous free surface. Consistent with the strip-theory long-wave assumption, the free surface in the vicinity of the section can be approximated as a flat plane passing through a point $\vec{X}_\zeta = (X_c, Y_c, \zeta_c)$ with the free surface unit normal $\vec{n}_w$ pointing out of water, where $\zeta_c = \zeta(t; X_c, Y_c)$.

Following this generalized definition of H, we can show that (see Appendix for derivation)

$$H = \frac{(\vec{X}_\zeta - \vec{X}_c) \cdot \vec{n}_w}{S} = \frac{\Delta Z_c n_{w,Z}}{S}, \tag{1}$$

where $\Delta Z_c = \zeta_c - Z_c$ and

$$S = \sqrt{1 - (\vec{n}_w \cdot \vec{k})^2} = \sqrt{1 - n_{w,Z}^2}. \tag{2}$$

Note that $n_{w,Z}$ and $n_{w,z}$ are the components of $\vec{n}_w$ in the earth-fixed vertical Z-direction and the section-local axial z-direction, respectively.

Another important quantity is $\dot{H}$, the time derivative of H. Instead of differentiating H numerically, $\dot{H}$ is more conveniently and reliably computed from the body velocity of the section at $\vec{X}_c$, $\vec{V}_b(t; \vec{X}_c)$, and the flow velocity on the free surface at $\vec{X}_\zeta$, $\vec{V}_f(t; \vec{X}_\zeta)$.

Throughout the present development, we shall consistently neglect the rotation of the member section about its center by treating $\dot{\vec{\imath}}$, $\dot{\vec{\jmath}}$, and $\dot{\vec{k}}$ as zeros. This is customary of strip-theory formulations. We will also neglect $\dot{\vec{n}}_w$ following the long-wave approximation. With these simplifications, it can be shown that (see Appendix for derivation)

$$\dot{H} = \frac{\left(\vec{V}_f(t; \vec{X}_\zeta) - \vec{V}_b(t; \vec{X}_c)\right) \cdot \vec{n}_w}{S}. \tag{3}$$

Note that Eq. (1) and Eq. (3) break down when the member axis is exactly normal to the free surface, resulting in $S = 0$. However, this does not present any issue because we only intend to apply the new model to members that are at a shallow angle from the free surface.

Without wave stretching, we always treat the SWL as the free surface. The expressions for H and $\dot{H}$ simplify to

$$H = \frac{Z_{\mathrm{SWL}} - Z_c}{\sqrt{1 - k_Z^2}}, \tag{4}$$

$$\dot{H} = \frac{-V_{b,Z}\left(t; \vec{X}_c\right)}{\sqrt{1 - k_Z^2}}. \tag{5}$$

Additionally, it is useful to have the area of the section that is submerged, $A_s(H)$:

$$A_s(H) = \begin{cases} 0 & \text{if } H < -R \\ R^2 \cos^{-1}\left(-\frac{H}{R}\right) + H\sqrt{R^2 - H^2} & \text{if } -R \le H \le R, \\ \pi R^2 & \text{if } H > R \end{cases} \tag{6}$$

with the derivative

$$\frac{\mathrm{d}A_s}{\mathrm{d}H} = \begin{cases} 0 & \text{if } H < -R \\ 2\sqrt{R^2 - H^2} & \text{if } -R \le H \le R. \\ 0 & \text{if } H > R \end{cases} \tag{7}$$

2.2 Distributed transverse loads

With strip theory, the total transverse hydrodynamic force (per unit length) on a member section is given by

$$\vec{F}_{tot} = \vec{F}_I + \vec{F}_A + \vec{F}_D, \tag{8}$$

where $\vec{F}_I$ is the fluid inertia component proportional to flow acceleration, $\vec{F}_A$ is the added-mass component proportional to body acceleration, and $\vec{F}_D$ is the quadratic drag force. $\vec{F}_I$ can be further separated into a Froude-Krylov component, $\vec{F}_{FK}$, and a scattering component, $\vec{F}_{Sc}$, which forms a duo with the added-mass component, $\vec{F}_A$. We can combine $\vec{F}_{Sc}$ and $\vec{F}_A$ into $\vec{F}_{dist}$, the force due to body disturbance of the inflow, leading to the following alternative force decomposition:

$$\vec{F}_{tot} = \vec{F}_{FK} + \vec{F}_{dist} + \vec{F}_D. \tag{9}$$

HydroDyn's existing semi-numerical hydrostatic load calculation based on pressure integration on the instantaneous wetted surface (MHstLMod=2) can already accommodate near-horizontal members on the free surface and will not be discussed further here. The hydrostatic contribution also includes a sectional moment. This is necessary for consistent hydrostatic restoring loads. For the hydrodynamic contributions, any sectional moment about the section center, $\vec{X}_c$, is neglected.

2.2.1 Force due to body disturbance of the flow

For a cylinder just beneath or breaching the free surface, the section added mass strongly depends on the center depth H given by Eq. (1). We have the section sway added mass $\mu_{xx}(H)$ and heave added mass $\mu_{yy}(H)$. We further assume $\mu_{xy} = \mu_{yx} = 0$. In general, $\mu_{xx}(H) \neq \mu_{yy}(H)$ for a surface-piercing section.

Let the in-plane body velocity be $\vec{v}_b = (u_b, v_b)$ and the flow velocity be $\vec{v}_f = (u_f, v_f)$. $\vec{v}_f$ is evaluated at the section center if submerged or on the free surface if the section center is above water. The relative velocity is $\vec{v}_r = \vec{v}_b - \vec{v}_f = (u_r, v_r)$. The first component is in the horizontal x-direction, and the second component is in the in-plane "vertical" direction.

The change in flow momentum due to body disturbance is given by

$$\vec{P}_{dist} = \mu_{xx}(H) u_r \vec{\imath} + \mu_{yy}(H) v_r \vec{\jmath}, \tag{10}$$

where $\vec{\imath}$ and $\vec{\jmath}$ are the unit vectors in the section local x- and y-directions. The hydrodynamic force on the body section (force/length) is given by

$$\vec{F}_{dist} = -\frac{\mathrm{d}\vec{P}_{dist}}{\mathrm{d}t} = -\left(\frac{\mathrm{d}\mu_{xx}}{\mathrm{d}H}\dot{H}u_r + \mu_{xx}\dot{u}_r\right)\vec{\imath} - \left(\frac{\mathrm{d}\mu_{yy}}{\mathrm{d}H}\dot{H}v_r + \mu_{yy}\dot{v}_r\right)\vec{\jmath}, \tag{11}$$

where $\dot{H}$ can be computed from the section velocity and flow velocity on the free surface using Eq. (3). Again, we are neglecting section rotation by treating $\vec{\imath}$ and $\vec{\jmath}$ as constant. Regrouping the terms recovers the standard added-mass contribution, wave scattering contribution, and an additional water-entry/exit contribution:

$$\vec{F}_{dist} = \underbrace{-\mu_{xx}\dot{u}_b\vec{\imath} - \mu_{yy}\dot{v}_b\vec{\jmath}}_{\text{Added mass}} + \underbrace{\mu_{xx}\dot{u}_f\vec{\imath} + \mu_{yy}\dot{v}_f\vec{\jmath}}_{\text{Scattering}} \underbrace{- \frac{\mathrm{d}\mu_{xx}}{\mathrm{d}H}\dot{H}u_r\vec{\imath} - \frac{\mathrm{d}\mu_{yy}}{\mathrm{d}H}\dot{H}v_r\vec{\jmath}}_{\text{Water entry \& exit}}. \tag{12}$$

This derivation based on momentum conservation is attributed to the work on slamming loads by von Kármán (1929) but generalized here to oblique water entry and exit. However, the main reason for including the water-entry/exit terms here is not to model slamming loads (more sophisticated models exist for that purpose) but to simply maintain momentum conservation as the added mass changes with submergence over time.

$\vec{F}_{dist}$ can be written in a more compact form as

$$\vec{F}_{dist} = -\mathbf{M}\dot{\vec{v}}_r - \dot{H}\frac{\mathrm{d}\mathbf{M}}{\mathrm{d}H}\vec{v}_r, \tag{13}$$

where

$$\mathbf{M} = \begin{bmatrix} \mu_{xx}(H) & 0 \\ 0 & \mu_{yy}(H) \end{bmatrix}. \tag{14}$$

Note that the continuity of $\vec{F}_{dist}$ with submergence depends on the continuity and smoothness of $\mathbf{M}(H)$. When $H = -R, \frac{\mathrm{d}\mathbf{M}}{\mathrm{d}H} \neq 0$ in general, so $\vec{F}_{dist}$ is not continuous at the instant of water contact. This is essentially the slamming load. This impulsive slamming load can cause difficulties with the structural solver; therefore, ad-hoc smoothing is required for numerical stability at the expense of physical fidelity.

As submergence increases, $\mu_{xx}(H)$ and $\mu_{yy}(H)$ both approach the deeply submerged isotropic added mass, μ, losing the dependence on submergence. As a result, $\vec{F}_{dist}$ reduces to the standard Morison equation added-mass and scattering terms:

$$\vec{F}_{dist} = -\mu\dot{\vec{v}}_r = -\rho\pi R^2\dot{\vec{v}}_r. \tag{15}$$

To compute $\vec{F}_{dist}$, $\mu_{xx}(H)$, $\mu_{yy}(H)$, and their derivatives with respect to H must be determined beforehand. Two options can be considered. With the first option, a simple engineering approximation can be used:

$$\mu_{xx}(H) = \mu_{yy}(H) = \frac{A_s(H)}{\pi R^2}\mu. \tag{16}$$

The submerged area of the section, $A_s(H)$, has continuous first derivative with respect to H; therefore, $\vec{F}_{dist}$ is automatically C^0. Equation (6) and Eq.(7) give $A_s(H)$ and $\frac{\mathrm{d}A_s(H)}{\mathrm{d}H}$. This simple engineering approximation is general and applies to any cross-section geometry; however, it lacks any physical or theoretical basis. Therefore, we opt for the second formulation below for circular sections.

As an alternative formulation, $\mu_{xx}(H)$ and $\mu_{yy}(H)$ can be estimated from the 2D (vertical plane) potential-flow solution of a horizontal cylinder that is just below or breaches the free surface. This is the formulation currently implemented. The potential-flow hydrodynamic coefficients for near-surface bodies are frequency dependent. This effect cannot be captured by strip-theory models that do not possess a radiation convolutional integral for wave radiation damping. Therefore, the choice of frequency requires some care.

For consistency, we should choose the added mass at either the low- or high-frequency limit, where radiation damping is typically zero. With the slamming problem, it is customary to use the infinite-frequency limit as opposed to the zero-frequency limit common with strip-theory models due to the short timescale involved. In this work, we are not planning to accurately resolve the slamming load but nevertheless use the infinite-frequency limit in the heave direction for a more practical reason. The heave added mass of a 2D surface-piercing body is logarithmically infinite at zero frequency with a non-zero wave radiation damping. This is simply incompatible with strip-theory models. Fortunately, the use of infinite-frequency added mass in the heave direction is also generally acceptable. We

consider two scenarios. In the first, the structure predominantly consists of horizontal slender members floating on the free surface. In this case, the structure will be mostly wave-following with resonance frequencies above typical wave frequencies due to high hydrostatic stiffness relative to body mass. In this case, the vertical-plane responses in waves, i.e., heave and pitch, of the structure are typically dominated by hydrostatic stiffness and the Froude-Krylov force with added mass playing a secondary role. In this case, using infinite-frequency added mass allows the model to more consistently capture the short-timescale loads due to nonlinear wetting. Note that the heave and pitch resonance frequencies can still be off due to the use of infinite-frequency added mass; however, for wave response, it is not critical to model these frequencies exactly as long as they are well above the wave frequencies. The error in added mass is also smaller at higher resonance frequencies. In the second scenario, the horizontal slender surface members only constitute a (small) part of the structure. With such structures, the global resonance frequencies are often dictated by other components of the structure, and some error in the added mass of the horizontal surface members can still be acceptable.

In the horizontal/sway direction, the picture is different. The horizontal-plane resonance frequencies for surge, sway, and yaw can be very low with soft mooring. Because of this, the model resonance frequency and the resulting drift motion can have large errors in the first scenario, where the structure predominantly consists of strip-theory members floating on the surface. To address this limitation, a formulation involving mixed frequency limits is adopted. The sway added-mass term uses μ_{xx} from the zero-frequency limit, which remains finite for 2D bodies, while the water-entry/exit term derived from instantaneous momentum conservation continues to use the more appropriate infinite-frequency value of $\frac{\mathrm{d}\mu_{xx}}{\mathrm{d}H}$. This is a compromise that delivers both more accurate slow drift response and physical short-timescale water-entry/exit forces at the expense of strict instantaneous momentum conservation.

The infinite-frequency added mass of a horizontal cylinder is available from Greenhow and Ahn (1988) as a function of $\bar{H} = H/R$. Figure 2 shows $\bar{\mu}_{xx}(\bar{H})$ and $\bar{\mu}_{yy}(\bar{H})$ based on either exact solution if available or fitted relations developed for this work when closed-form exact solutions are not available. The infinite-frequency sway and heave added mass are both C^1 for $H > -R$, although the heave added mass has a jump in slope at $H = -R$, the point of tangent contact, generating the impulsive slamming force.

For a surface-piercing circular section ($-1 \leq \bar{H} < 1$), the infinite-frequency heave added mass as a function of submergence is given by Taylor (1930):

$$\bar{\mu}_{yy} = \frac{\mu_{yy}}{\rho\pi R^2} = \frac{\pi^2}{3}\frac{1-\cos\theta}{(2\pi-\theta)^2} + \frac{1-\cos\theta}{6} + \frac{\sin\theta-\theta}{2\pi}, \tag{17}$$

where

$$\theta = 2\cos^{-1}(-\bar{H}). \tag{18}$$

Equation (17) breaks down as $\bar{H} \to 1$. In the immediate vicinity of this point, we can rewrite Eq. (17) in terms of $\phi = 2\pi - \theta$ and take the Taylor expansion about $\phi = 0$, leading to the following asymptotic form that can be used instead:

$$\begin{aligned}\bar{\mu}_{yy} = &\left(\frac{\pi^2}{6} - 1\right) + \left(\frac{1}{12} - \frac{\pi^2}{72}\right)(2\pi-\theta)^2 + \frac{1}{12\pi}(2\pi-\theta)^3 \\ &+ \left(\frac{\pi^2}{2160} - \frac{1}{144}\right)(2\pi-\theta)^4 + O(\phi^5).\end{aligned} \tag{19}$$

There are no exact closed-form solutions for the sway and heave added mass for full submergence ($\bar{H} \geq 1$) (Greenhow and Li, 1987). Therefore, suitable monotonic polynomial approximations are developed by satisfying key end conditions. For the fully submerged regime, heave and sway added mass are the same and are described by the same quintic polynomial approximation. This approximation is developed by satisfying the known value, $\frac{\pi^2}{6} - 1$, and slope, $\frac{\pi^2}{9} - \frac{2}{3}$, at $\bar{H} = 1$ and a truncated deep-submergence limit at $\bar{H} = 5$, where we require the added mass to be exactly unity with zero slope. Monotonicity is also enforced to fully determine the coefficients.

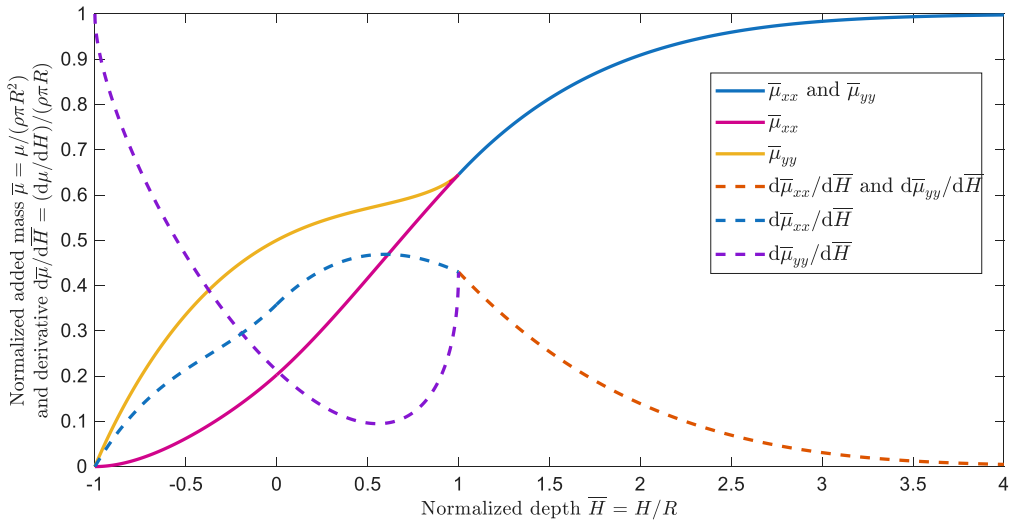


Figure 2. Infinite-frequency sway (x) and heave (y) added mass of a circular section in the vertical plane in the vicinity of a free surface. The solid and colored lines are the exact expression or fitted polynomials of the added mass as functions of the normalized center depth, $\bar{H}$. The dashed lines are the derivatives with respect to $\bar{H}$ based on either the exact expression or fitted polynomials.

Although not currently used, we also provide a convenient polynomial approximation of the infinite-frequency sway added mass for future reference. For sway added mass with partial submergence, a two-piece quartic polynomial approximation is developed with a boundary at $\bar{H} = 0$. For the section of $-1 \le \bar{H} \le 0$, the polynomial approximation is developed by enforcing monotonicity and the boundary conditions of $\bar{\mu}_{xx} = 0$ and zero slope at $\bar{H} = -1$ and $\bar{\mu}_{xx} = \frac{2}{\pi^2}$ at $\bar{H} = 0$, a known exact value. The second piece over $0 \le \bar{H} \le 1$ also satisfies monotonicity, the same boundary value at $\bar{H} = 0$ and the known value and slope at $\bar{H} = 1$. Finally, we choose the slope at $\bar{H} = 0$ by requiring C^2 at this point, leading to a slope of 0.3583. The polynomial coefficients are summarized in Table 1.

Table 1. Polynomial coefficients that approximate infinite-frequency added mass

	Heave and sway $1 \le \bar{H} \le 5$	Sway only $-1 \le \bar{H} \le 0$	Sway only $0 \le \bar{H} \le 1$
$\bar{H}^5$	3.286758827989542E-04	0	0
$\bar{H}^4$	-8.289169840152171E-03	8.957417595114259E-02	1.791483519022852E-02
$\bar{H}^3$	8.303624394187833E-02	1.321603211375043E-01	-1.321603211375040E-01
$\bar{H}^2$	-4.153257652497481E-01	1.982404817062564E-01	1.982404817062560E-01
$\bar{H}^1$	1.043012143185959E+00	3.582967038045702E-01	3.582967038045703E-01
$\bar{H}^0$	-5.782806107250993E-02	2.026423672846756E-01	2.026423672846756E-01

To compute the section loads, we also need the derivatives of μ_{xx} and μ_{yy} with respect to H, which can be straightforwardly obtained. When $-1 \le \bar{H} < 1$, the infinite-frequency heave added mass can be differentiated with respect to $\bar{H}$ as

$$\begin{aligned}\frac{\mathrm{d}\bar{\mu}_{yy}}{\mathrm{d}\bar{H}} &= \left(\frac{\pi^2}{3}\frac{\sin\theta}{(2\pi-\theta)^2} + \frac{2\pi^2}{3}\frac{1-\cos\theta}{(2\pi-\theta)^3} + \frac{\sin\theta}{6} + \frac{\cos\theta - 1}{2\pi}\right)\frac{2}{\sqrt{1-\bar{H}^2}} \\ &= -\frac{4\pi^2}{3}\frac{\bar{H}}{(2\pi-\theta)^2} + \frac{8\pi^2}{3}\frac{\sin\frac{\theta}{2}}{(2\pi-\theta)^3} - \frac{2}{3}\bar{H} - \frac{2}{\pi}\sin\frac{\theta}{2}.\end{aligned} \tag{20}$$

Equation (20) breaks down as $\bar{H} \to 1$. At this limit, we have the following asymptotic form that can be used instead, again by the Taylor expansion about $\phi = 2\pi - \theta = 0$:

$$\frac{\mathrm{d}\bar{\mu}_{yy}}{\mathrm{d}\bar{H}} = \left(\frac{\pi^2}{9} - \frac{2}{3}\right) - \frac{1}{\pi}(2\pi - \theta) + \left(\frac{1}{12} - \frac{\pi^2}{360}\right)(2\pi-\theta)^2 + O(\phi^3) \text{ as } \bar{H} \to 1. \tag{21}$$

The polynomial approximations can be directly differentiated to obtain the coefficients in Table 2. The dimensional added-mass derivatives are simply

$$\frac{\mathrm{d}\mu}{\mathrm{d}H} = \rho\pi R\frac{d\bar{\mu}}{d\bar{H}}. \tag{22}$$

Table 2. Polynomial coefficients that approximate the derivatives of added mass

	Heave and sway $1 \le \bar{H} \le 5$	Sway only $-1 \le \bar{H} \le 0$	Sway only $0 \le \bar{H} \le 1$
$\bar{H}^4$	1.643379413994771E-03	0	0
$\bar{H}^3$	-3.315667936060868E-02	3.582967038045703E-01	7.165934076091407E-02
$\bar{H}^2$	2.491087318256350E-01	3.964809634125129E-01	-3.964809634125119E-01
$\bar{H}^1$	-8.306515304994961E-01	3.964809634125128E-01	3.964809634125119E-01
$\bar{H}^0$	1.043012143185959E+00	3.582967038045702E-01	3.582967038045703E-01

Lastly, we require the zero-frequency sway added mass shown in Figure 3. The zero-frequency sway added mass is nonmonotonic in $\bar{H}$ with a slope discontinuity at $\bar{H} = 1$. Strictly speaking, this is acceptable because we do not use the derivative of the zero-frequency added mass for the water-entry/exit force in the mixed-frequency-limit formulation. Nevertheless, we replace the exact solution with a smoothed curve that removes the cusp at $\bar{H} = 1$ to prevent any unforeseen numerical difficulties caused by a non-smooth change in added mass.

For the partially submerged regime ($-1 \le \bar{H} < 1$), we have the analytical solution from conformal mappings by Taylor (1930):

$$\bar{\mu}_{xx} = \frac{\sin\theta - \theta}{2\pi} + \frac{\theta}{6}(1 - \cos\theta)\frac{4\pi - \theta}{(2\pi - \theta)^2}, \tag{23}$$

where θ is still given by Eq. (18). In the present implementation, we only use Eq. (23) up to $\bar{H} = 0.8$. For $0.8 < \bar{H} \le 1.1$, we use a quintic polynomial in $\bar{H} - 1$ to remove the cusp. For $1.1 < \bar{H} \le 5$, we use a quartic polynomial in $1/\bar{H}$ that smoothly converges to $\bar{\mu}_{xx} = 1$ at $\bar{H} = 5$, effectively truncating the shallow-submergence tail. For $\bar{H} > 5$, we have a constant $\bar{\mu}_{xx} = 1$. The required coefficients are summarized in Table 3.

Table 3. Polynomial coefficients providing smoothed zero-frequency sway added mass

$0.8 < \bar{H} \le 1.1$		$1.1 < \bar{H} \le 5$	
Term	Coefficient	Term	Coefficient
$(\bar{H} - 1)^5$	3.136836992349601E+03	$\bar{H}^{-5}$	0
$(\bar{H} - 1)^4$	1.319449187374385E+03	$\bar{H}^{-4}$	6.909412843781649E+00
$(\bar{H} - 1)^3$	5.558658150878864E+01	$\bar{H}^{-3}$	−1.337160281428654E+01
$(\bar{H} - 1)^2$	−3.919588885535995E+01	$\bar{H}^{-2}$	9.714661186777143E+00
$(\bar{H} - 1)^1$	−2.965948375729567E+00	$\bar{H}^{-1}$	−2.502373347997483E+00
$(\bar{H} - 1)^0$	2.104159421957775E+00	$\bar{H}^0$	1.207805984092653E+00

With the mixed-frequency-limit formulation, the derivative of the zero-frequency sway added mass is not required and is not provided here.

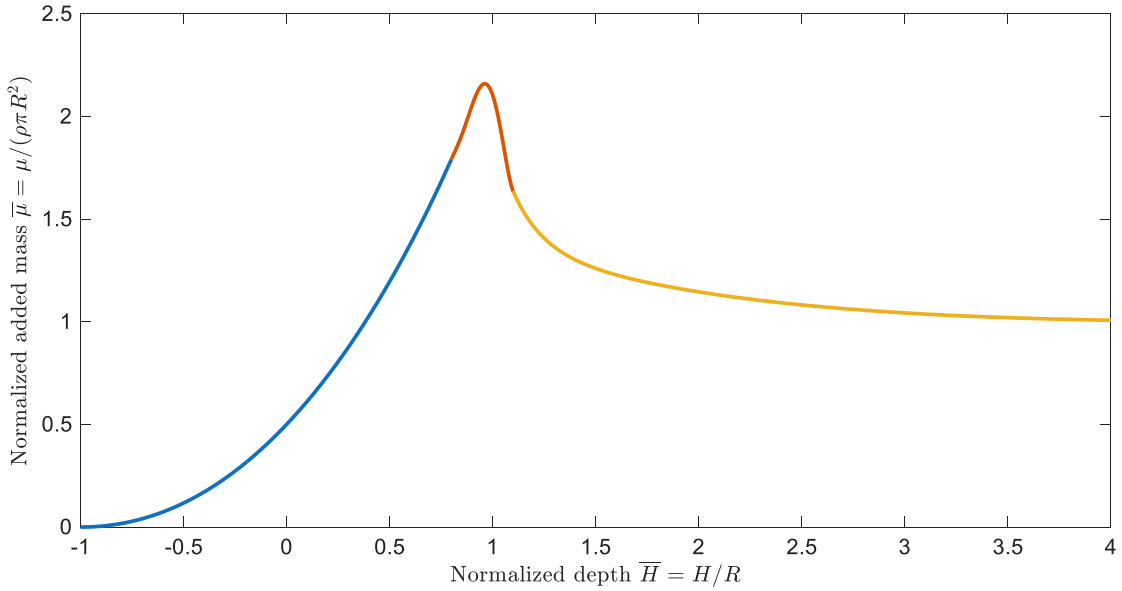


Figure 3. Zero-frequency sway added mass of a circular section in the vertical plane in the vicinity of a free surface. The colored lines are the exact expression or fitted curves of the added mass as a function of the normalized center depth, $\overline{H}$. The fitted curve intentionally removes the cusp at $\overline{H} = 1$.

The sway and heave added mass implemented and their derivatives are shown in Figure 4 and Figure 5, respectively, with comparison to the simple engineering approximation based on $A_s(H)/(\pi R^2)$. The significant differences between the area-based approximation and the true potential-flow solution motivated the use of the latter.

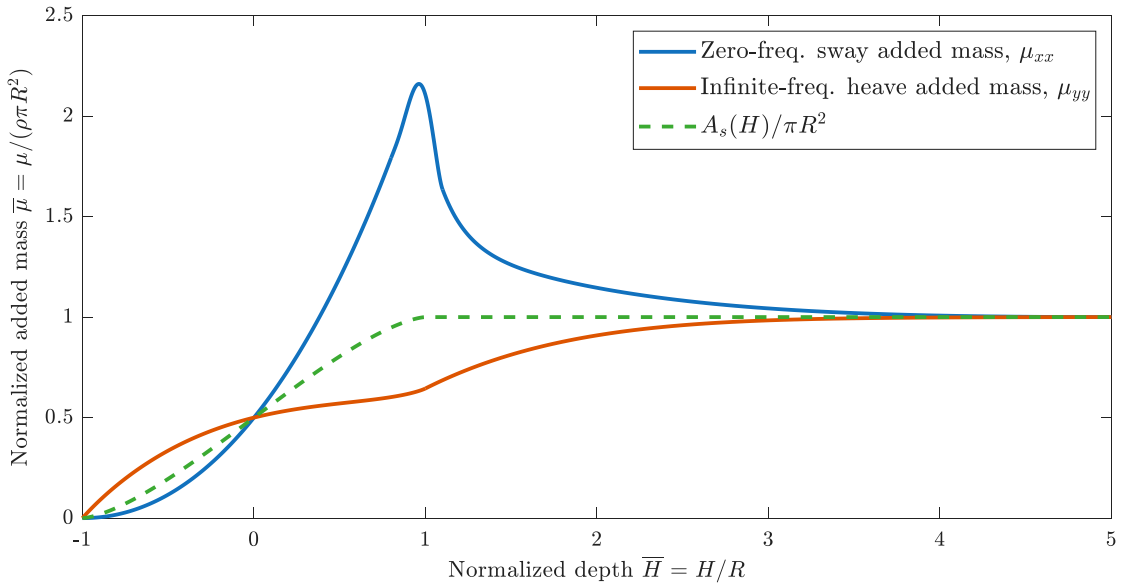


Figure 4. The submergence-dependent section sway and heave added mass implemented with comparison to the simple engineering approximation based on submerged-area scaling.

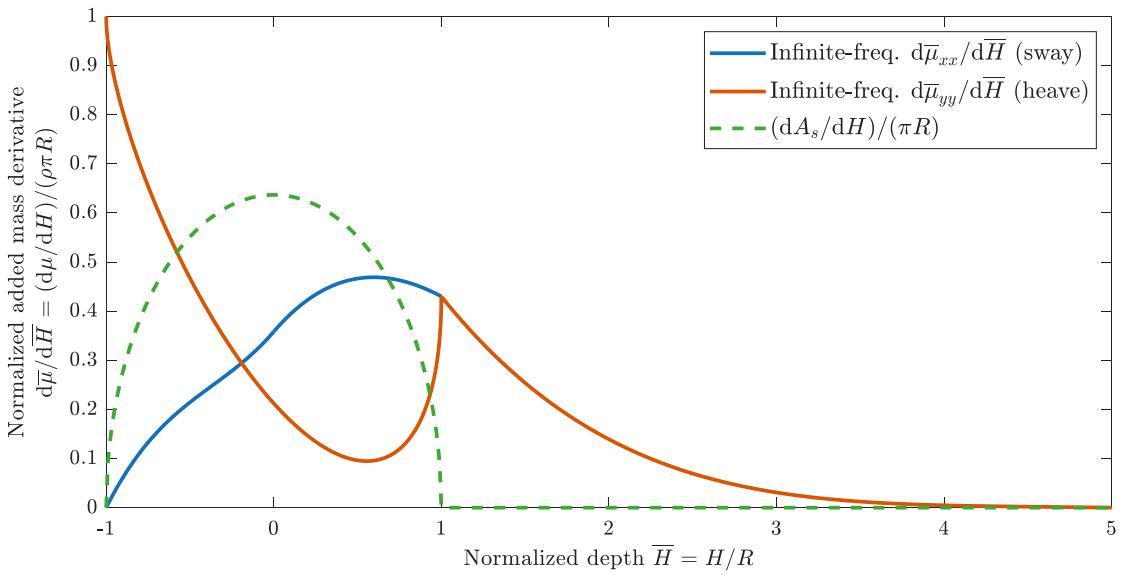


Figure 5. The derivatives of the submergence-dependent section sway and heave added mass implemented with comparison to the simple engineering approximation based on submerged-area scaling.

In HydroDyn, the user can also specify an isotropic added-mass coefficient, C_A, for a circular member. We shall interpret this as the deep-submergence added-mass coefficient and simply multiply C_A to $\mu_{xx}(H)$ and $\mu_{yy}(H)$ obtained above.

2.2.2 Froude-Krylov force

From the SeaState wave module, we can query the inflow hydrodynamic pressure, p_{dyn}, and acceleration, $\vec{a}_f$, at the section center, $\vec{X}_c$. If $\zeta_c \geq Z_c$, i.e., the center is submerged, SeaState will return $p_{dyn}(\vec{X}_c)$ and $\vec{a}_f(\vec{X}_c)$ directly. Alternatively, if the center is out of water, the hydrodynamic pressure and flow acceleration at $\vec{X}_\zeta$ can be queried. For conciseness, let the wavefield variables at $\vec{X}_0$ be known with

$$\vec{X}_0 = \begin{cases} \vec{X}_c & \text{if } \zeta_c \geq Z_c \\ \vec{X}_\zeta & \text{otherwise} \end{cases}. \tag{24}$$

With potential-flow theory, we have

$$\nabla p_{dyn} = -\rho \vec{a}_f. \tag{25}$$

Note that for linear wave theory, $\vec{a}$ is simplified to the Eulerian acceleration. For high-order wave theory, $\vec{a}$ should be the material acceleration. The pressure at a point on the perimeter of the circular section

$$\vec{X}_R = \vec{X}_c + R\vec{r} \tag{26}$$

is, to leading order, simply

$$\begin{aligned} p_{dyn}(\vec{X}_R) &= p_{dyn}(\vec{X}_0) + \nabla p_{dyn}(\vec{X}_0) \cdot (\vec{X}_c + R\vec{r} - \vec{X}_0) \\ &= p_{dyn}(\vec{X}_0) + \nabla p_{dyn}(\vec{X}_0) \cdot (\vec{X}_c - \vec{X}_0) + \nabla p_{dyn}(\vec{X}_0) \cdot (R\vec{r}) \\ &= p_{dyn}(\vec{X}_c) - \rho R \vec{a}_f(\vec{X}_0) \cdot \vec{r}, \end{aligned} \tag{27}$$

where $\vec{r}$ is the unit radial vector from $\vec{X}_c$ to $\vec{X}_R$:

$$\vec{r} = \vec{\imath}\cos\theta + \vec{\jmath}\sin\theta. \tag{28}$$

The Froude-Krylov force on the section (force/length) is given by

$$\vec{F}_{FK} = -\int_{\theta_1}^{\theta_2} p_{dyn}\vec{r}R \, \mathrm{d}\theta, \tag{29}$$

where $\theta_1 \to \theta_2$ is the wetted side of the circular section. We intentionally replace the wall normal vector, $\vec{n}$, with the radial unit vector, $\vec{r}$, effectively neglecting the contribution from any wall tapering. This matches the existing formulation for submerged members, where the axial force due to tapering is computed separately. Using the expression for p_{dyn} and carrying out the integration, we obtain

$$\begin{aligned} F_{FK,x} = -p_{dyn}(\vec{X}_c)R(\sin\theta_2 - \sin\theta_1) + \rho R^2 a_{f,x}\left(\frac{\theta_2 - \theta_1}{2} + \frac{\sin 2\theta_2 - \sin 2\theta_1}{4}\right) \\ + \rho R^2 a_{f,y}\frac{\cos 2\theta_1 - \cos 2\theta_2}{4} \end{aligned} \tag{30}$$

$$\begin{aligned} F_{FK,y} = p_{dyn}(\vec{X}_c)R(\cos\theta_2 - \cos\theta_1) + \rho R^2 a_{f,x}\frac{\cos 2\theta_1 - \cos 2\theta_2}{4} \\ + \rho R^2 a_{f,y}\left(\frac{\theta_2 - \theta_1}{2} - \frac{\sin 2\theta_2 - \sin 2\theta_1}{4}\right) \end{aligned} \tag{31}$$

If the section is fully submerged, the above expression reduces to the standard Morison equation Froude-Krylov term:

$$\vec{F}_{FK} = \rho\pi R^2 \vec{a}_f(\vec{X}_c). \tag{32}$$

This also guarantees the force is continuous as the section transitions from partial submergence to full submergence. For consistency with existing implementation, we allow the user to scale $\vec{F}_{FK}$ by the user-defined pressure coefficient, C_P.

2.2.3 Drag force

The drag force on a surface-piercing section is extremely complex. As an engineering model, we simply assume the following general form:

$$\vec{F}_D = -\frac{1}{2}\rho C_D(H) d(H) |\vec{v}_r| \vec{v}_r, \tag{33}$$

where

$$d(H) = \begin{cases} 0 & \text{if } H < -R \\ 2\sqrt{R^2 - H^2} & \text{if } -R \le H < 0 \\ 2R & \text{if } H \ge 0 \end{cases}. \tag{34}$$

The exact form of $C_D(H)$ is left open for now. Initial implementation shall assume a constant C_D. $\vec{F}_D$ is C^0 at the point of $H = -R$.

2.3 Distributed axial loads on tapered members

Similar to the distributed transverse loads, the hydrodynamic distributed forces in the axial direction can be expressed as

$$\vec{F}_{tot,ax} = \vec{F}_{FK,ax} + \vec{F}_{dist,ax} + \vec{F}_{D,ax} \tag{35}$$

The total 6DOF hydrostatic force and moment are computed together and will not be discussed here.

2.3.1 Force due to body disturbance of the flow

The distributed axial loads along a tapered member should also be derived consistently based on momentum conservation. The momentum change of the fluid due to body disturbance in the axial direction is given by

$$\vec{P}_{dist,ax} = \mu_{zz} w_r \vec{k} = \mu_{zz}(w_b - w_f)\vec{k}, \tag{36}$$

where μ_{zz} is the section axial added mass, w_b is the body axial velocity, and w_f is fluid velocity in the axial direction at the section center or on the free surface if the center is above water.

The axial force on the tapered section due to body disturbance of the flow is given by:

$$\vec{F}_{dist,ax} = -\frac{\mathrm{d}\vec{P}_{dist,ax}}{\mathrm{d}t} = -\mu_{zz}\dot{w}_r\vec{k} - \dot{\mu}_{zz} w_r \vec{k}. \tag{37}$$

Again, $\dot{\vec{k}}$ is neglected. Expanding the expression leads to the same three components observed in the transverse direction:

$$\vec{F}_{dist,ax} = \underbrace{-\mu_{zz}\dot{w}_b\vec{k}}_{\text{Added mass}} + \underbrace{\mu_{zz}\dot{w}_f\vec{k}}_{\text{Scattering}} - \underbrace{\frac{\mathrm{d}\mu_{zz}}{\mathrm{d}H}\dot{H} w_r \vec{k}}_{\text{Water entry \& exit}}. \tag{38}$$

To support partially submerged sections, we can scale the fully submerged section axial added-mass force by the fraction, f_c, of the circumference that is wetted:

$$\mu_{zz} = 2C_{Aax}\rho\pi R^2 \left|\frac{\mathrm{d}R}{\mathrm{d}\ell}\right| f_c(H), \tag{39}$$

where C_{Aax} is the user-defined axial added-mass coefficient and

$$f_c(H) = \begin{cases} 0 & \text{if } H < -R \\ \frac{1}{\pi}\cos^{-1}\left(-\frac{H}{R}\right) & \text{if } -R \le H \le R. \\ 1 & \text{if } H > R \end{cases} \tag{40}$$

This leads to

$$\frac{\mathrm{d}\mu_{zz}}{\mathrm{d}H} = \begin{cases} 0 & \text{if } H < -R \\ \frac{2C_{Aax}\rho R^2}{\sqrt{R^2 - H^2}}\left|\frac{\mathrm{d}R}{\mathrm{d}\ell}\right| & \text{if } -R \le H \le R, \\ 0 & \text{if } H > R \end{cases} \tag{41}$$

which poses an issue at the point of tangent contact with $H = \pm R$ where $\frac{\mathrm{d}\mu_{zz}}{\mathrm{d}H} \to \infty$ like $\frac{1}{\sqrt{|H|-R}}$. To address this issue, smoothing is again required near the points of tangent contact between the section and the free surface.

2.3.2 Froude-Krylov force

The axial Froude-Krylov component is also straightforwardly generalized to partially wetted sections using f_c as

$$\vec{F}_{FK,ax} = 2C_{Pax}p_{dyn}\pi R\frac{\mathrm{d}R}{\mathrm{d}\ell}f_c(H)\vec{k}, \tag{42}$$

where C_{Pax} is the user-defined pressure coefficient. Unlike $\vec{F}_{dist,ax}$, $\vec{F}_{FK,ax}$ does not require the derivative $f_c(H)$; therefore, smoothing is not required.

2.3.3 Drag force

Finally, the axial drag force is generalized as

$$\vec{F}_{D,ax} = -\frac{1}{2}\rho C_{Dax}\pi R\left|\frac{\mathrm{d}R}{\mathrm{d}\ell}\right| f_c(H)|w_r|w_r\vec{k}, \tag{43}$$

where the user-defined axial drag coefficient, C_{Dax}, can potentially be made a function of H as well but treated as a constant in this initial implementation.

2.4 Joint loads

Finally, we have endplate loads at the joints. Unlike hydrostatic loads, joint hydrodynamic loads are computed on a per-joint basis instead of a per-member basis. Each joint has a characteristic directional area, $\vec{A}_J = A_J\vec{n}_A$, and directional volume, $\vec{V}_J = V_J\vec{n}_V$, estimated by summing the directional contributions from all members connecting to the joint. For the purpose of computing hydrodynamic loads, we effectively treat each joint as a circle of area A_J and normal $\vec{n}_A$ or as a hemisphere with volume V_J and equatorial plane normal $\vec{n}_V$ depending on the load component. The effective radius of the endplate is given by

$$R = \sqrt{A_J/\pi}. \tag{44}$$

The hydrodynamic force on the joint due to body disturbance has a familiar form:

$$\vec{F}_{dist,J} = \underbrace{-\mu_J \dot{v}_{b,n}\vec{n}_V}_{\text{Added mass}} + \underbrace{\mu_J \dot{v}_{f,n}\vec{n}_V}_{\text{Scattering}} - \underbrace{\frac{\mathrm{d}\mu_J}{\mathrm{d}H}\dot{H}v_{r,n}\vec{n}_V}_{\text{Water entry \& exit}}. \tag{45}$$

The endplate added mass, generalized to partially wetted endplates, is given by

$$\mu_J = \rho V_J C_{JA}\left(\frac{A_s(H)}{A_J}\right)^{3/2}. \tag{46}$$

The "slamming" coefficient follows:

$$\frac{\mathrm{d}\mu_J}{\mathrm{d}H} = \frac{3}{2}\rho V_J C_{JA}\frac{\sqrt{A_s(H)}}{{A_J}^{3/2}}\frac{\mathrm{d}A_s}{\mathrm{d}H}. \tag{47}$$

Unlike the distributed axial added mass for tapered members, $\frac{\mathrm{d}\mu_J}{\mathrm{d}H}$ is zero at both $H = -R$ and $H = R$, so smoothing is not required.

Finally, both the Froude-Krylov and drag terms can be straightforwardly generalized to

$$\vec{F}_{FK,J} = -C_{JP}p_{dyn}\frac{A_s(H)}{A_J}\vec{A}_J, \tag{48}$$

$$\vec{F}_{D,J} = \frac{A_s(h)}{A_J}\vec{F}_{D,J0}. \tag{49}$$

There are multiple formulations for the fully submerged endplate drag force, $\vec{F}_{D,J0}$, in HydroDyn, including velocity high-pass filtering and the suction-side-only formulation, but the above simple area-based scaling can be applied to any model. In the actual implementation, we simply scale the filtered and unfiltered velocity by $\sqrt{A_s/A_J}$ without changing the expressions for drag force. This way, $\vec{F}_{D,J}$ is automatically scaled by A_s/A_J.

Another subtle change to the implementation is the location where the necessary wave kinematics and dynamic pressure is evaluated. Previously, these quantities were always evaluated at the joint (endplate center) and were zeroed if the joint was out of water. With the modified implementation, these quantities are evaluated at the joint if the joint is in water. If the joint is out of water, the quantities on the free surface below the joint are evaluated and never zeroed. As the endplate moves out of water, the joint loads are scaled by the submerged area ratio and are automatically zeroed if fully out of water.

3 Additional implementation details

3.1 Load smoothing

We have two force components that are either discontinuous or singular at tangent contact between the circular section and the free surface. These are the distributed water-entry/exit force in the heave direction and in the axial direction.

For numerical stability, we need to smooth out these forces. A simple cubic smoothing function can be used:

$$w(s) = \begin{cases} 0 & \text{if } s < 0 \\ 3s^2 - 2s^3 & \text{if } 0 \le s \le 1. \\ 1 & \text{if } s > 1 \end{cases} \tag{50}$$

For the transverse slamming heave force:

$$-\frac{\mathrm{d}\mu_{yy}}{\mathrm{d}H}\dot{H}v_r\vec{j}, \tag{51}$$

we only need to apply smoothing as $H \to -R$ through multiplication with a smoothing factor $w(s_1)$, where

$$s_1 = \frac{R + H}{\epsilon R}. \tag{52}$$

The choice of the smoothing length ϵR presents a compromise between physical fidelity and load smoothness. We choose $\epsilon = 0.2$ for smoothness (10% of the section diameter). This is acceptable because we do not intend to capture the impulsive slamming force exactly.

For the axial slamming force:

$$-\frac{\mathrm{d}\mu_{zz}}{\mathrm{d}H}\dot{H}w_r\vec{k}, \tag{53}$$

we need to apply smoothing when H is close to $\pm R$ through multiplication with a smoothing factor $w(s_1)w(s_2)$, where

$$s_2 = \frac{R - H}{\epsilon R}. \tag{54}$$

Note that $w(s) \to 0$ quadratically as $s \to 0$. This is adequate for removing the $\frac{1}{\sqrt{|H|-R}}$ singularity present with $\frac{\mathrm{d}\mu_{zz}}{\mathrm{d}H}$.

3.2 Blending of load models

The new distributed hydrodynamic load model is intended to work in parallel with the existing model for members at a steeper angle relative to the free surface. We need to smoothly blend the loads from the two models based on the section inclination.

Let $\vec{F}_1$ be the total hydrodynamic force at a member node computed using the existing model and let $\vec{F}_2$ be the force computed for the same node using the new model. The final blended force is given by

$$\vec{F}_{\text{blended}} = w(s)\vec{F}_1 + \big(1 - w(s)\big)\vec{F}_2, \tag{55}$$

with

$$s = \frac{\left|\vec{n}_w \cdot \vec{k}\right| - \sin(25°)}{\sin(35°) - \sin(25°)}. \tag{56}$$

To take into account local wave slope and potential member deflection, the force blending is performed on a nodal basis, instead of on a member basis.

If $s < 0$ or $s > 1$, only one of the two forces, $\vec{F}_1$ and $\vec{F}_2$, needs to be computed. The unused model can be skipped. This improves performance and avoids numerical difficulties, such as Eq. (1) and Eq. (3) breaking down for a member perpendicular to the free surface.

This integrated and blended framework should allow the new hydrodynamic strip-theory model to automatically take effect when appropriate without requiring additional user inputs.

For the joint loads, the new generalized formulations outlined in Section 2.4 work for both fully and partially submerged endplates. Therefore, the same set of formulations is used in most cases without the need for any blending. The only exception happens when the endplate is almost parallel to the free surface, i.e., the endplate normal direction is within 5 degrees from the local free surface normal. In this case, Eq. (1) and Eq. (3) both break down, and we revert to the existing binary wetting. If the center of the endplate is above the free surface or the still water level, depending on the use of wave stretching, the endplate is treated as either fully dry or fully submerged with $\dot{H} = 0$. In essence, we do not and cannot model the slamming force on a plate parallel to the free surface.

3.3 Input file changes

Currently, there are no changes to HydroDyn's input file. With the model blending for distributed loads and the unified model for joint loads, the new model will automatically take effect when appropriate for cylindrical members. If the structure undergoes large

rotation and a member approximately aligned with the free surface becomes more vertical, the new model will gradually phase out in favor of the existing model.

However, if we choose to grant the user finer control over the model behavior, we can add more inputs to the HydroDyn input file in the future, including allowing the user to set the slamming load smoothing distance as a fraction of section radius/diameter, ϵ, and/or the load blending angle ranges. New input will also be required if we want to enable submergence-dependent drag coefficients. These changes will be deferred from this initial implementation in favor of an unmodified input file.

4 References

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5 Appendix

The derivation of Eq. (1) and Eq. (3) is included here for the record.

Let $\vec{X}_l$ be a point on the in-plane waterline cut. The submergence H is given by

$$H = \left(\vec{X}_l - \vec{X}_c\right) \cdot \vec{n}_l,$$

where $\vec{n}_l$ is the in-plane unit vector normal to the waterline cut. It is the projection of the actual free-surface normal, $\vec{n}_w$, onto the section plane:

$$\vec{n}_l = \frac{\vec{n}_w - \left(\vec{n}_w \cdot \vec{k}\right)\vec{k}}{\left|\vec{n}_w - \left(\vec{n}_w \cdot \vec{k}\right)\vec{k}\right|} = \frac{\vec{n}_w - \left(\vec{n}_w \cdot \vec{k}\right)\vec{k}}{\sqrt{1 - \left(\vec{n}_w \cdot \vec{k}\right)^2}}.$$

Since $\vec{X}_l$ is both on the planarized free surface and in the section plane, we have

$$\left(\vec{X}_l - \vec{X}_\zeta\right) \cdot \vec{n}_w = 0,$$

$$\left(\vec{X}_l - \vec{X}_c\right) \cdot \vec{k} = 0.$$

Substituting the expression for $\vec{n}_l$ and the two constraints on $\vec{X}_l$ above into the expression for H yields Eq. (1).

To derive Eq. (3), we need to differentiate Eq. (1) with respect to time while neglecting the time derivatives of $\vec{\imath}$, $\vec{\jmath}$, $\vec{k}$, and $\vec{n}_w$. This leads to

$$\dot{H} = \left(\dot{\zeta}_c - \dot{Z}_c\right)\frac{n_{w,Z}}{S},$$

where

$$\dot{\zeta}_c = \left[\frac{\partial \zeta}{\partial t} + V_{b,X}\frac{\partial \zeta}{\partial X} + V_{b,Y}\frac{\partial \zeta}{\partial Y}\right]_{t;X_c,Y_c},$$

$$\dot{Z}_c = V_{b,Z}.$$

The kinematic free-surface boundary condition at $\vec{X}_\zeta$ is

$$\left.\frac{\partial \zeta}{\partial t}\right|_{t;X_c,Y_c} = V_{f,Z}\left(t;\vec{X}_\zeta\right) - V_{f,X}\left(t;\vec{X}_\zeta\right)\left.\frac{\partial \zeta}{\partial X}\right|_{t;X_c,Y_c} - V_{f,Y}\left(t;\vec{X}_\zeta\right)\left.\frac{\partial \zeta}{\partial Y}\right|_{t;X_c,Y_c}.$$

Substituting the kinematic boundary condition into the expression for $\dot{\zeta}_c$ leads to

$$\dot{\zeta}_c - \dot{Z}_c = \left(V_{b,X} - V_{f,X}\left(t;\vec{X}_\zeta\right)\right)\left.\frac{\partial \zeta}{\partial X}\right|_{t;X_c,Y_c} + \left(V_{b,Y} - V_{f,Y}\left(t;\vec{X}_\zeta\right)\right)\left.\frac{\partial \zeta}{\partial Y}\right|_{t;X_c,Y_c} - \left(V_{b,Z} - V_{f,Z}\left(t;\vec{X}_\zeta\right)\right).$$

Finally, recall

$$\frac{\vec{n}_w}{n_{w,Z}} = \left(-\frac{\partial \zeta}{\partial X}, -\frac{\partial \zeta}{\partial Y}, 1\right).$$

Therefore,

$$\dot{H} = \frac{n_{w,Z}}{S}\left(\dot{\zeta}_c - \dot{Z}_c\right) = \frac{n_{w,Z}}{S}\left(\vec{V}_b - \vec{V}_f(t; \vec{X}_\zeta)\right) \cdot \left(-\frac{\vec{n}_w}{n_{w,Z}}\right)_{t;X_c,Y_c}$$
$$= \frac{1}{S}\left(\vec{V}_f\left(t; \vec{X}_\zeta\right) - \vec{V}_b\left(t; \vec{X}_c\right)\right) \cdot \vec{n}_w(t; X_c, Y_c).$$